Osman Akar

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Education

PhD in Applied Mathematics at University of California, Los Angeles 2019 - 2024 (exp.)
Research Interests: Machine Learning, Computational Linear Algebra

Masters in Mathematics at University of California, Los Angeles 2016 - 2019
Departmental Scholars Program (simultaneous BS and MA degrees awarded)

BS in Mathematics of Computation at University of California, Los Angeles 2015 - 2019
Math Undergraduate Merit Scholar, Graduated Cum Laude

Work Experience & Research

Research Intern at UE Physics Dev Team at Epic Games Los Angeles, CA
Machine Learning based Signed Distance Function *Jun 2022 - Present*

- Designed a novel learning-based method to compute signed distance function (SDF) of human body motion depending on the rotation of joints in the skeleton, so that the deformation of the skin/muscle (especially near the joints) looks as real as possible.
- Our method, called ML-SDF, is a combination of multiple ML models in which each represents different parts of the human body that involves a joint (e.g. elbow, shoulder, spine, legs, etc.).
- ML-SDF is used for character animation and its interaction with clothing in Unreal Engine. The method reduces the memory cost on the RAM significantly (less than 1% memory requirement compared to the actual levelsets it's replacing). The goal of the project is to automatize the whole process given the body mesh and its skeleton (e.g. different characters of Fortnite). This includes joint-based body part selection, dataset generation for the corresponding body parts, and the training, all to be done inside Unreal Engine.

Graduate Researcher at Teran Lab Los Angeles, CA
A Deep Conjugate Direction Method (DCDM) for Iteratively Solving Linear Systems *Feb 2021 - May 2022*

- Created CNN-based linear system solver, a modified version of the classical Conjugate Gradients (CG) method, for the systems coming from computational fluid dynamics applications (e.g. incompressible flows).
- Designed the network architecture and its custom loss function to perform unsupervised learning to speed up the solver. At the moment DCDM is 3.2 times faster than classical methods for the linear systems over 56 millions of degrees of freedom.
- Developped a novel method for creating the dataset so that it does not require any simulation input, as opposed to our competitors. This reduced the dataset size significantly, and consequently the time required to generate it and the training.
- The work is accepted and presented at **ICML 2023** at Hawaii. See https://icml.cc/virtual/2023/poster/24899

Teaching at UCLA Mathematics Department

- **Instructor** of *Introduction the Probability* class of 40 students at 23 Spring.
- **Lead of the Olympiads Instructor Group** in Olga Radko Math Circle (ORMC) for 21-22 and 22-23 school years.

List of Honors and Awards

- Putnam Performance Prize by UCLA Mathematics Department, 2019
- Honorable Mention at William Lowell Putnam Mathematical Competition, 2018. Ranked 77 among 4623 students.
- Richard F. Arens Putnam Scholars Undergraduate Award by UCLA Mathematics Department, 2016
- Sole recipient of UCLA Math Undergraduate Merit Scholarship in class of '19. Since its foundation at 2010, there are only 8 recipients of this scholarship all over the world.
- **Gold Medal** at International Mathematical Olympiad (**IMO**), 2014, South Africa
- Gold Medal with perfect score at Balkan Mathematical Olympiad (**BMO**), 2014, Bulgaria
- Silver Medal at International Mathematical Olympiad (**IMO**), 2013, Colombia

Programming & Technical

Languages: C++, Python, Matlab, LaTeX